

SAMPATH MYLAVARAPU

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SUMMARY

Data Engineer with **5+ years** of experience in designing scalable **ETL pipelines** and cloud data architectures. Proficient in **PySpark, Airflow, AWS, and Azure**. Specialized in optimizing data workflows for performance and cost efficiency. Hands-on experience with **AI/ML integration** for intelligent automation and driving actionable business insights.

EXPERIENCE

pSolv | Data Engineer

Aug 2024 - Present

- Developed a scalable AWS-based data pipeline leveraging **AWS Lambda, AWS Batch, and PySpark** to process **20+TB** of Machine-Readable Files (MRF) from **insurers**, enabling fault-tolerant data ingestion.
- Optimized data processing workflows by migrating **Spark** jobs from **EC2** to **AWS EMR**, converting raw JSON files into partitioned **Parquet** format stored in **S3**, reducing storage costs by **60%** and improving performance.
- Implemented ETL pipelines orchestration using **Apache Airflow** and **Linux shell scripting**, automating data workflows, scheduling tasks, and ensuring seamless execution.
- Managed **version control and CI/CD** using **GitHub**, ensuring code quality through peer reviews and automation.
- Built **PySpark** workflows to clean and enrich large datasets, and used **dbt** to model and load data into **AWS Redshift**, boosting query performance by **30%**.
- Built an **AI agent** utilizing **enterprise RAG** (retrieval-augmented generation) pipelines with **FAISS** to retrieve context for **LLM** and generate SQL queries, seamlessly integrating with **D3.js** dashboards for **instant visual insights**.

Arzoo | Data Engineer

Jun 2021 - Jun 2023

- Engineered scalable ETL pipelines using **PySpark, SparkSQL** to move data from Azure Data Lake Storage (ADLS) to **Azure Synapse Analytics**, following the medallion architecture, enabling faster analytics on **1M+** transactions.
- Led the migration of orchestration logic from Azure Data Factory (ADF) to **Databricks Workflows**, reducing cloud operational costs by **25%** and improving pipeline reliability and execution time.
- Designed and implemented a **Kimball-style dimensional data warehouse** with support for **SCDs** to enable historical tracking of product and vendor attributes, powering sales and inventory analytics across **50K+** SKUs.
- Migrated pipeline monitoring from **ADF to Grafana**, an open-source visualization tool, to reduce operational costs and enable real-time insights into ETL health and performance.
- Contributed to **CI/CD workflows** by troubleshooting production issues, implementing **hotfixes**, and deploying updates to maintain pipeline stability, data accuracy, and SLA compliance.

UBS | Analyst

Jul 2019 - May 2021

- Automated the end-to-end account creation process using **Python**, reducing manual effort by **70%** and cutting average **processing time by 50%**, ensuring faster and error-free on-boarding.
- Designed and optimized **SQL** queries for **SSRS reports**, implementing **CTEs**, window functions, and indexing strategies to improve report generation speed by **40%** and eliminate data inconsistencies in executive-level reporting.
- Preprocessed and cleaned data for use in **machine learning classification** algorithm that routed transactions to appropriate approval workflows based on type, boosting straight-through processing efficiency by **25%**.
- Developed **10+ Power BI dashboards** for executives to track key **KPIs** such as account creation time, transaction approval turnaround, and process accuracy, improving operational visibility and decision-making.

TECHNICAL SKILLS

Languages/Databases: Python, PySpark, SQL, SparkSQL, R Programming, MySQL, NoSQL, PostgreSQL, Oracle SQL

Frameworks/Tools: Apache Spark, Airflow, DBT, Power BI, Tableau, Grafana, Hadoop, Hive, Impala, Flume, SAS

Machine Learning & AI: NLP, Neural Networks, LLMs, RAG process, Random Forest, Regression, Classification.

Cloud & Data Platforms: ADLS, ADF, Azure Synapse, AWS S3, AWS Redshift, AWS EMR, Snowflake, Databricks

Certifications: [AWS Machine Learning Associate](#), [Microsoft Fabric Analytics Engineer Associate](#)

EDUCATION

The University of Texas at Dallas

Aug 2023 – May 2025

Master of Science in Business Analytics and Artificial Intelligence

GPA - 4.0

Birla Institute of Technology and Science, Pilani

Aug 2017 – Jun 2021

Bachelor of Engineering in Mechanical Engineering, Finance

GPA - 3.7

ACADEMIC PROJECTS

AI Powered Data Quality Fixes in ETL Pipelines

Jan 2025 - Present

- * Automated **data quality (DQ)** validation and correction using the **Qwen 32B LLM** (Ollama), enabling schema-aware anomaly detection, intelligent fix suggestions, semantic validations, and human-in-the-loop approval to improve data integrity, accuracy, and reduce manual overhead.

ELT Architecture with Snowflake, dbt, and Airflow

Jan 2025 - Apr 2025

- * Built an **ELT pipeline** to load and transform data in **Snowflake** using **dbt**, creating source, staging tables, and transformed models (fact tables, data marts) for optimized data processing. Orchestrated the entire process with **Airflow**, automating dbt model deployment and ensuring data quality through tests and efficient workflows for seamless integration.